

DATA QUALITY & EXPLORATORY DATA ANALYSIS REPORT

Telco Customer Churn Prediction

Project Name: Telco Customer Churn Prediction

Report Type: Data Quality and Exploratory Data Analysis (EDA)

Dataset: Telco Customer Churn

Dataset Source: <https://www.kaggle.com/datasets/blastchar/telco-customer-churn>

Prepared By: chhaya kumari

Date: 9 October 2026

1. Introduction

The purpose of this report is to examine the Telco Customer Churn dataset, identify data quality issues, and explore customer characteristics that may be associated with churn.

The analysis will help us understand the dataset before preparing it for machine learning. It will also help identify potential factors that a telecommunications company can investigate to improve customer retention.

2. Dataset Overview

The Telco Customer Churn dataset contains customer demographics, account information, subscribed services, billing details, and customer churn status.

Attribute	Description
Number of rows	7,043
Number of columns	21
Target variable	Churn
Target values	Yes, No
Problem type	Binary classification
Data format	CSV

The target variable **Churn** indicates whether a customer has left the telecommunications service.

3. Data Quality Assessment

The dataset must be checked for missing values, duplicate records, incorrect data types, and inconsistent category values.

3.1 Missing Values

The **TotalCharges** column contains blank-string entries that need to be converted into missing values before analysis.

Recommended action: Convert **TotalCharges** to a numeric data type, identify missing values, and decide how to handle them based on further investigation.

3.2 Duplicate Records

Duplicate rows should be checked to determine whether repeated records exist.

Recommended action: Identify duplicate rows and investigate them before deciding whether removal is appropriate.

3.3 Data Types

Some columns contain numerical information stored as text. In particular, **TotalCharges** should be converted to a numeric data type.

Recommended action: Validate the data type of each column and convert columns where necessary.

3.4 Categorical Values

Certain service columns contain values such as **No internet service** and **No phone service**. These values represent service conditions and should not automatically be treated as missing data.

Recommended action: Inspect the categories and preserve their meaning during preprocessing.

4. Exploratory Data Analysis

EDA will be used to understand customer behavior and identify patterns related to churn.

4.1 Overall Churn Distribution

The **Churn** column will be analyzed to calculate the proportion of customers who left the service compared with those who stayed.

Business relevance: Understanding the churn rate helps establish a baseline for the prediction problem.

4.2 Churn by Contract Type

Customer churn will be compared across contract categories, such as month-to-month, one-year, and two-year contracts.

Business relevance: This analysis can help determine whether churn differs across contract types.

4.3 Churn by Tenure

Customer tenure represents the length of time a customer has remained with the company.

Business relevance: Comparing tenure with churn may reveal whether newer customers behave differently from long-term customers.

4.4 Churn by Payment Method

Churn will be examined across the available payment methods.

Business relevance: This may identify customer groups that warrant further investigation into billing or payment experiences.

4.5 Churn by Tech Support

Churn will be compared between customers with and without technical support services.

Business relevance: This can help investigate whether service support is associated with customer retention.

5. Initial Business Hypotheses

The following hypotheses will be tested using the dataset. They are not confirmed findings at this stage.

1. Customers with month-to-month contracts may have higher churn.
2. Customers with shorter tenure may be more likely to leave.
3. Churn may vary across payment methods.
4. Customers without technical support may have different churn rates.
5. Customer service subscriptions and billing characteristics may be associated with churn.

These hypotheses must be validated through data analysis before drawing conclusions.

6. Data Cleaning Plan

The following steps are planned before model development:

1. Load the dataset and inspect its dimensions.
2. Check for missing values and duplicate rows.
3. Convert **TotalCharges** to numeric format.
4. Investigate records with missing **TotalCharges**.
5. Validate categorical values.
6. Review numerical distributions and potential outliers.
7. Document all cleaning decisions.
8. Prepare the cleaned dataset for feature engineering.

7. Limitations

- The dataset represents a particular telecommunications customer population and may not generalize to every company.
- Observed relationships do not necessarily establish causation.
- Missing or inconsistent values may affect the results.
- Business hypotheses must be tested using actual analysis.
- Predictive performance cannot be determined until machine learning models are trained and evaluated.

8. Expected Outcomes

After completing EDA and data quality checks, the project is expected to produce:

- A documented assessment of dataset quality.
- An understanding of the distribution of churn.
- Visual comparisons of churn across customer characteristics.
- A list of validated or rejected business hypotheses.
- A cleaned dataset suitable for subsequent machine learning tasks.

9. Conclusion

The Telco Customer Churn dataset provides information that can be used to study customer retention. Before building a predictive model, it is important to understand the data structure, resolve data quality issues, and investigate relationships between customer attributes and churn.

The next phase will focus on completing the EDA notebook, documenting the actual findings, and preparing the data for machine learning.